

KCPC Week 2 Problems

Advanced Invariants, Heuristics, and Algorithmic Bridges

How To Use This Sheet

Each section builds on Week 1 techniques. Try the problems in order; later ones generalise earlier ideas. For algorithmic bridge problems, sketch the algorithm and argue correctness before writing code.

1 Advanced Logic And Common Knowledge

Basic – Blue-Eyed Islanders ($k = 2$)

There are 2 islanders with blue eyes, and the rest have brown. They cannot see their own eyes. A trusted outsider announces: “At least one of you has blue eyes.” Each day, anyone who knows their own eye colour leaves at midnight. What happens and on which day?

Easy – Cheating Husbands

In a village of 100 couples, some husbands have been unfaithful. Every wife knows which *other* husbands have cheated, but not her own. The queen announces: “At least one husband has been unfaithful.” If a wife deduces her husband cheated, she must act at midnight. What happens?

Medium – Sum And Product Puzzle

Two integers x, y with $2 \leq x < y$ and $x + y \leq 100$. Mr P knows the product xy ; Mr S knows the sum $x + y$. P: “I don’t know the numbers.” S: “I knew you didn’t know.” P: “Now I know the numbers.” S: “Now I know them too.” Find x and y .

Hard – Cheryl’s Birthday

Cheryl gives Albert and Bernard a list of 10 possible dates. She whispers the month to Albert and the day to Bernard. Albert: “I don’t know the birthday, but I know Bernard doesn’t know either.” Bernard: “At first I didn’t know, but now I do.” Albert: “Now I also know.” When is Cheryl’s birthday?

Very Hard – 100 Prisoners And A Light Bulb

100 prisoners are in solitary cells. There is a central room with a light bulb (initially off) and a switch. One prisoner at a time is chosen randomly to visit the room; they may toggle the switch and/or announce “everyone has visited.” If correct, all go free; if wrong, all die. They may agree a strategy beforehand. Design a protocol that guarantees freedom.

2 Advanced Invariants And Monovariants

Basic – Peg Solitaire (Small Board)

On a 3×3 board with the centre empty, pegs jump orthogonally over adjacent pegs into empty squares, removing the jumped peg. Can you finish with one peg in the centre?

Easy – Token Game Modulo 3

Tokens on integer points. A move: choose x with tokens at x and $x + 1$; remove them and place one at $x + 2$. Start with one token at 0 and one at 1. Can you ever have a token at 100? (Hint: track $\sum 2^{\text{position}}$ modulo 3.)

Medium – Conway’s Soldiers

On an infinite checkerboard, pegs occupy all squares with $y \leq 0$. A peg jumps orthogonally over another into an empty square, removing the jumped peg. Can a peg ever reach $y = 5$? (Hint: assign weights $x^{|y|}$ with $x = (\sqrt{5} - 1)/2$.)

Hard – Polynomial Invariant For Grid Walks

A token starts at $(0, 0)$. A move changes (x, y) to $(x+1, y+2)$ or $(x+2, y+1)$. Characterise all reachable positions. (Hint: find a polynomial $P(x, y)$ invariant under both moves.)

Very Hard – IMO 1986 Problem 3 (Sequence Invariant)

Given integer sequence $a_0, a_1, \dots$ with $a_0 = 0$, $a_1 = 1$, and $a_{n+1} = a_n + a_{n-1} + 1$. Prove that $a_n^2 - a_{n-1}a_{n+1}$ is invariant modulo 3, and find its value.

3 Colouring And Tiling: Generalisations

Basic – Tromino Tiling (Deficient Board)

A $2^n \times 2^n$ board has one square removed. Prove it can be tiled by L-shaped trominoes (three squares in an L). (Inductive construction.)

Easy – Tiling With T-Tetrominoes

Can a 4×4 board be tiled by T-tetrominoes (four squares in a T shape)? Can a $4 \times n$ board for $n > 4$? Use a 4-colouring based on coordinates mod 2.

Medium – Fault-Free Domino Tiling

A domino tiling of a rectangle has a fault line if a straight line from one side to the other cuts no dominoes. Prove that a 6×6 board has a fault-free tiling, but a 6×8 board does not.

Hard – Triominoes On A Chessboard

Can an 8×8 chessboard with one corner removed be tiled by straight trominoes (1×3 rectangles)? Use a 3-colouring of the board.

Very Hard – Generalised Mutilated Chessboard

For which m, n can an $m \times n$ board with two squares of opposite colours removed be tiled by dominoes? Prove your characterisation.

4 Induction And Recursion: From Proof To Algorithm

Basic – Josephus Problem

n people stand in a circle. Eliminate every second person until one remains. Where should you stand to survive? Find a recurrence and closed form for $J(n)$.

Easy – Gray Codes

An n -bit Gray code lists all 2^n binary strings so consecutive strings differ in one bit. Give a recursive construction. Prove it works by induction.

Medium – Recursive 15-Puzzle Parity

Given a 4×4 15-puzzle configuration, design a recursive algorithm to determine if it is solvable. Use the invariant from Week 1.

Hard – Counterfeit Coin With n Weighings

Generalise the 12-coin, 3-weighing problem. With n weighings on a balance scale, what is the maximum number of coins among which you can find one counterfeit (heavy or light)? Give a strategy achieving the bound.

Very Hard – Hadamard Matrix Construction

A Hadamard matrix of order n is an $n \times n$ matrix with entries ± 1 whose rows are mutually orthogonal. Give a recursive construction for $n = 2^k$ and prove it works.

5 Zeitz Heuristics In Depth

Basic – Symmetry Strategy In Pairing Game

Two players alternately place a domino on a 4×4 board. Dominoes cannot overlap. The player who cannot move loses. Show the second player has a winning strategy by symmetry.

Easy – Extreme Principle For Points In A Circle

Given 5 points in a unit circle, prove that some pair is at distance ≤ 1 . (Consider the extreme point farthest from the centre.)

Medium – Wishful Thinking For A Number Puzzle

Find all integer solutions to $x^2 + y^2 = 2z^2$ with $x, y, z > 0$. (Wishful thinking: assume x, y, z form a Pythagorean-like structure; reduce to a known form.)

Hard – Reduction To Graph Reachability

n lamps in a circle, each on or off. A move toggles lamp i and its two neighbours. From a given configuration, can you reach all off? Model as a graph and use linear algebra over $\mathbb{F}_2$.

Very Hard – Contest Problem: Combine Heuristics

(From a past olympiad) A finite set of integers has the property that for any two distinct elements, their sum is also in the set. Prove the set is either $\{0\}$ or contains at least two elements whose sum is 0. Use symmetry, extreme principle, and reduction.

6 Algorithmic Bridges

Basic – Water Jugs As Graph Reachability

Two jugs of capacities 3 and 5. Allowed: fill, empty, pour between them. Model as a graph with states (a, b) . Use BFS conceptually to find a sequence to measure exactly 4 units.

Easy – Greedy Activity Selection

You have n intervals $[s_i, f_i]$. Choose a maximum-size subset of non-overlapping intervals. Prove that picking the interval with earliest finish time, then recursing, is optimal. (Exchange argument.)

Medium – DP From Induction: Coin Change

Given coin values $c_1, \dots, c_k$ and target n , find the minimum number of coins to make exactly n . Define $f(n)$ recursively, identify overlapping subproblems, and give a tabulation algorithm.

Hard – Sliding Puzzle State-Space Search

2×3 sliding puzzle with tiles 1–5 and one empty. From a given start state, find the minimum moves to reach the goal. Describe the state encoding, transitions, and why BFS works. (You may assume at most 181,440 reachable states.)

Very Hard – Greedy Scheduling With Deadlines

n jobs, each takes 1 unit of time, has a deadline d_i and profit p_i . You may schedule at most one job per time slot. A job only earns profit if completed by its deadline. Design a greedy algorithm to maximise total profit and prove it correct. (Hint: consider jobs in decreasing profit order; use a disjoint-set union to find latest free slot.)

